Table 7.3: Difference between Capsule and Slime layer

Capsule	Slime layer
Capsule is a glycocalyx layer, consisting of firmly associated polysaccharide molecules with the cell wall.	Slime layer is a glycocalyx layer that consists of loosely associated glycoprotein molecules.
It is a well-organized layer, difficult to be washed off.	It is an unorganized layer and can be easily washed off.
It is tightly bound to the cell wall.	It is loosely bound to the cell wall.
It is thicker than slime layer.	It is a thin glycocalyx layer.
It acts as a virulence factor that helps to escape phagocytosis.	It mainly helps in adherence. It protects the cell from dehydration and nutrient loss.

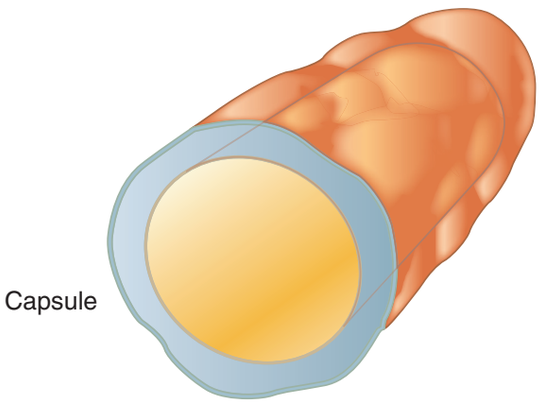


Figure 7.6: Structure of capsule

The role of the capsule varies depending on the bacterium.

- A thick capsule protects cells from dehydration.
- Capsules protect the pathogenic bacteria from being engulfed and destroyed by white blood cells (phagocytes).
- Capsules are virulence factors of many pathogenic bacteria, such as *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Bacillus anthracis*. Encapsulated bacterial cells generally have greater virulence.

Slime layer

Some bacteria are covered with a surface layer that is loosely distributed around the cell and diffuses into the medium, this surface layer is referred to as slime layer. (Figure 7.7) The slime layer is a structure that is easily washed off. Slime layer protects bacteria from loss of water and nutrients. Slime has little affinity for basic dyes and is ivisible in Gram stained smears.

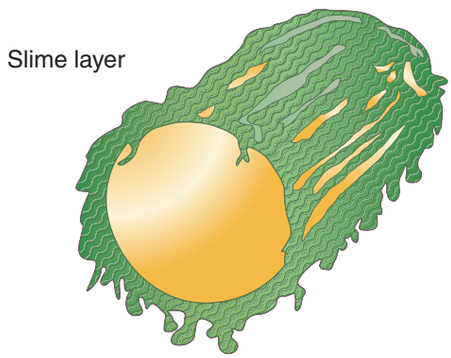


Figure 7.7: Structure of slime layer

7.2.3 Other Appendages

Sheath

Sheathed bacteria are bacteria that grow as long filaments in the form of chain or trichome. These bacteria are enclosed by a hollow tube like structure known as sheath

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